AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

T216B
Cisco Networking (CCCNA) part B

Final Examination
Spring Semester 2025/2026

Date: 8/06/2026

Number of Exam Pages: 5 (including this cover sheet)	**Time Allowed:** 3h

Instructions:

- Total Marks: **50**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part 1: Multiple choices. 1 mark per question. **Total: 6 Marks**

1) _____ is used to perform host verification and DoS attacks.
   a. ICMP echo request and echo reply
   b. ICMP unreachable
   c. ICMP mask reply
   d. ICMP router discovery
2) A company is expanding its business to other countries. All branch offices must remain connected to corporate headquarters at all times. Which network technology is required to support this scenario?
   a. WLAN
   b. WAN
   c. MAN
   d. LAN
3) Under which condition does congestion occur on a converged network with voice, video, and data traffic?
   a. If the request for bandwidth exceeds the amount of bandwidth available
   b. If video traffic requests more bandwidth than voice traffic requests
   c. If voice traffic latency begins to decrease across the network
   d. If a user downloads a file that exceeds the file limitation that is set on the server
4) Why would a network administrator issue the **show cdp neighbors** command on a router?
   a. To display the router ID and other information about OSPF neighbors
   b. To display device ID and other information about directly connected Cisco devices
   c. To display routing table and other information about directly connected Cisco devices.
   d. To display line status and other information about directly connected Cisco devices
5) What is an SNMP management agent?
   a. A database that a device keeps about network performance
   b. A computer loaded with management software and used by an administrator to monitor a network
   c. Software that is installed on devices managed by SNMP
   d. A communication protocol that is used by SNMP
6) A company uses a cloud-based payroll system. Which cloud computing technology is the company using?
   a. Browser as a service (Baas)
   b. Infrastructure as a service (IaaS)
   c. Software as service (SaaS)
   d. Wireless as a service (WaaS)

Part 2: Answer all the questions – 4 marks per question **Total: 20 Marks**

1) Answer the following questions:
   a) What is a floating static route? (1 mark)
   b) What is a stub router? (1 mark)
   c) What is a full specified static route? (1 mark)
   d) Mention one reason where dynamic routing is better to be used than static routing. (1 mark)
2) Answer the following questions:
   a) Mention one advantage of using multiarea OSPF? (1 mark)

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b) In a multiaccess network, the OSPF elects a DR. On which base a router is elected as a DR? (1 mark)
c) In a multiaccess network, what will happen if the DR fails? (1 mark)
d) A router has three physical interfaces and two loopback interfaces Each of these interfaces have been assigned an IP address as follow:
   - G0/1: 190.160.20.150
   - G0/2: 200.15.20.5
   - G1/1: 56.15.20.200
   - Lo1: 172.10.9.15
   - Lo2: 90.15.70.111

   What will be the Router ID of this router? (1 mark)

3) Answer the following questions:
   a) Math the security terms to their appropriate description. (Not all the terms will be used). (2 marks)

   Security terms
   - Risk
   - Vulnerability
   - Threat
   - Asset
   - Exploit
   - Mitigation

	Description	Security Term
A	_____ is the counter-measure that reduces the likelihood or severity of a potential threat or risk.	
B	_____ is the likelihood of a threat to exploit the vulnerability of an asset, with the aim of negatively affecting an organization	
C	_____ is anything of value to the organization. It includes people, equipment, resources, and data.	
D	_____ is a weakness in a system, or its design, that could be exploited by a threat.	

   b) What is the purpose of Denial of Service (DoS) attack? (1 mark)
   c) Why do threat actors use reconnaissance attacks?" (1 mark)

4) Answer the following questions:
   a) Why is there a need to have at least one permit ACE statement in an access control list (ACL)? (1 mark)
   b) Where should a standard ACL be placed? (1 mark)
   c) At which layer(s) of the OSI reference models does the extended ACL operates? (1 mark)
   d) In Cisco IOS, what number ranges are used to identify standard ACLs? (1 mark)

5) Answer the following questions:
   a) A company has 50 PCs connected to a LAN using private IP addressing. The connection to the internet is limited; and the network administrator has determined that only 10 PCs should be connected at the same time to the internet. What type of NAT should be used? (1 mark)
   b) Mention one advantage of NAT. (1 mark)
   c) Which type of VPN should be used in the following scenario: users are always on the move and do not have a fixed location but need to securely connect to the enterprise. (1 mark)
   d) List two advantages of VPN (1 mark)

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Part 3: Answer all the following questions **Total: 24 Marks**

1) Configure OSPF. (10 marks)

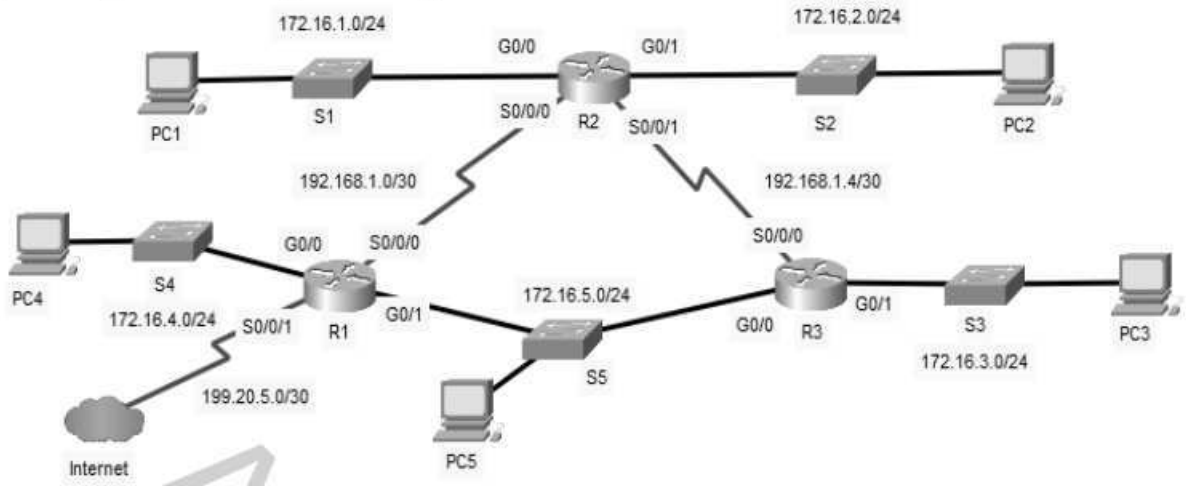


Figure 1

Scenario

OSPF was configured on all routers except R1. All the directly connected networks on these networks were advertised.

A) Determine the number networks that are directly connected to R1 and the number of remote networks for R1? (2 marks)
B) List all the directly connected networks to R1 (the prefix (network address) and the prefix length). (1 marks)

*** ***For the remaining questions you will need to write the commands*** ***

C) Configure a default static route on R1 to route all unmatched traffic via interface S0/0/1 (1 mark)
D) Enable OSPF on router R1 (refer to figure 1) (1 mark)
Note: the process ID is 85
E) Advertise all the directly connected networks of router R1. (Refer to figure 1) (3 marks)
F) Disable the advertisement of OSPF on R1 Interfaces G0/0 and G0/1(1 mark)
G) Advertise the default static route to all other routers. (1 mark)

2) Answer the following questions (14 marks)
A) Having received the following syslog message (figure 2), determine:
- Severity
- MNEMONIC
- Facility
- Message (2 marks)

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Figure 2

B) From the subnet mask 255.255.240.0 determine the equivalent wildcard mask (note: show the steps of calculation) (2 marks)
C) The network Administrator wants to configure an Extended ACL on router R3. Complete the configuration based on the following information: (4 marks)

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- Allow http traffic from network 192.168.1.0/24 to network 192.168.3.0/24
- Allow all traffic from IP address 192.168.2.10 to all destination
- Allow ftp traffic from network 172.16.4.0/26 to server 192.168.3.10
- Allow https traffic from network 172.16.5.0/27 to all destination
- Deny all other traffic
- Apply the ACL on traffic entering interface G0/2

```
Access-list 123 permit TCP 192.168.1.0 0.0.0.255 ____  0.0.0.255 eq  ____
Access-list 123 permit  ____  host 192.168.2.10 any
Access-list 123 permit TCP 172.16.4.0  0.0.0.63  ____  192.168.3.10 eq ftp
Access-list 123 permit TCP 172.16.5.0 ____  any eq https
Access-list 123 IP  ____  any any
____  G0/2
Ip ____123 in
```

Figure 3

D) The network administrator is configuring **NAT overload** on router R1. Complete the following configuration based on the following information: (3 marks)

- 5 public IP addresses will be used (from 188.10.23.5 to 188.10.23.9)
- NAT is allowed for network 192.168.0.0/16
- The inside network is on interface S0/1/1 and the outside network is on interface S0/1/0

```
ip nat pool Finance 188.10.23.5  _____  netmask 255.255.255.0
 _____  15 permit 192.168.0.0 0.0.255.255
ip nat inside source list  _____  pool  _____
interface serial 0/1/1
ip nat  _____
interface serial 0/1/0
ip nat  _____
```

Figure 4

E) Write a standard access control list which allows only an IP address (192.168.5.10) from network 192.168.5.0/24 and allows all other traffic. Three commands are needed to complete this task. (3 marks)

End of questions

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